MIL-C-60654A(MU)
24 September 1971
SUPERSEDING
MIL-C-60654(MU)
12 September 1967

MILITARY SPECIFICATION
CARTRIDGE, IMPULSE, M57 ASSEMBLY

1. SCOPE

1.1 This specification covers Cartridge, Impulse, M57, for use in the indoctrination of aircrew personnel.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military
MIL-A-2550 - Ammunition and Special Weapons, General Specification for

STANDARDS

Military
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions

DRAWINGS

Munitions Command
D8594084 - Cartridge, Impulse, M57 Assembly
C8796526 - Box, Packing, Ammunition, for Cartridge, Impulse, M57 in Metal Container, M21

FSC 1377

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2.1 (Cont'd)

F8796529 - Ammunition Container, Metal M21 for Cartridge, Impulse M57
IEL 8594084 - Inspection Equipment List for Cartridge Impulse M57 Assembly

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity as directed by the contracting officer.)

2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

ASA B46.1 - Surface Texture

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, N.Y. 10018)

3. REQUIREMENTS

3.1 General.-Materials, parts and assemblies shall conform to the applicable drawings and referenced specifications.

3.2 Functional.

3.2.1 The cartridge shall not fail to fire.

3.2.2 Ballistics.-The cartridge when assembled in an M6 Catapult shall propel a 300 pound mass vertically and shall meet the requirements of Table I.

TABLE I

Temperature Degrees (°) Fahrenheit (F)	Minimum Velocity Feet per Second (ft/sec)	Maximum Acceleration (gs)	Max. Rate of Change of Acceleration (gs/sec)	Max. Vertical Height of Travel (feet)
70°F ± 5°F	20	12	150	17.4

3.3 Workmanship.-Workmanship shall be in accordance with the requirements of the applicable drawings and specifications.

3.3.1 Metal defects.-The metal part or assembly shall be free from cracks, splits, cold shuts, inclusions, porosity or other metal defects.

3.3.2 Burr.-The part shall not have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

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3.3.3 Foreign matter.-No part or assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facility suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.2 Inspection.-Inspection shall be in accordance with MIL-A-2550.

4.2 First article sample.

4.2.1 Initial production sample.-At the beginning of regular production, a sample shall be submitted in accordance with contract requirements. The sample shall consist of 5 metal part assemblies and 10 sets of unassembled components for visual and diminsional examination and tests. The sample shall be manufactured using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.-After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Ballistics.-Twenty-one samples for ballistics shall be temperature conditioned at 70°F ± 5°F and ballistically tested for compliance with the requirements of Table I.

4.2.1.3 Initial production sample failure.-Failure of the sample to comply with requirements of the drawings and specifications shall result in sample disapproval.

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4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submittal of product.--The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Size of lot.--An ammunition lot shall consist of an identifiable group of metal part assemblies consisting of:

a. Components produced by a homogeneous process

b. Propellant from not more than one lot

c. Igniter from not more than one lot

d. Primers from not more than one lot

4.3.2 Examination.--One hundred percent examination shall be performed for all critical defects. Examination for major or minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in Tables II and III. For an individual defect (single characteristic) the criteria in Table II shall be used. For total defects within a major or minor class, the criteria in Table III shall be used. All non-conforming material shall be rejected. The roughness comparison specimens prescribed in ASA B46.1 shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

TABLE II

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination	-	-	-	-
301 - 1300	110	1	2	2	3
1301 - 3200	150	2	3	3	4
3201 - 8000	225	3	4	4	5
8001 - 22000	300	4	5	5	6

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4.3.2 (Cont'd)

TABLE III

Total Allowable Defects Per Sample Size

	Major		Minor	
	Accept	Reject	Accept	Reject
Sample Size for Major Defects	110	150	225	300
Number of Defects in Class				
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
over 43	6	8	11	14
Sample Size for Minor Defects	110	150	225	300
Number of Defects in Class				
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
over 48	8	11	17	20

4.3.2.1 Primer, Percussion, Drawing D10551686, covering a detail of Drawing D8594084.

Defect	Method of Inspection
CRITICAL	
1. Primer foil discolored	Visual
2. Mixed primers	Visual
MAJOR	
101. Burr	Visual
102. Foreign matter	Visual
MINOR	
None defined	

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4.3.2.2 Case, Cartridge, Drawing F8594085, covering a detail of Drawing D8594084.

Defect	Method of Inspection
CRITICAL	
1. Metal defective	Visual
MAJOR	
101. Concentricity of head with body at datum locations	Gage
102. Concentricity of primer pocket flash hole with primer pocket	Gage
103. Concentricity of primer pocket with body at datum locations	Gage
104. Concentricity of flange with body at datum locations	Gage
105. Diameter of body at two locations, min.	Gage
106. Diameter inside at counterbore end	Gage
107. Depth of primer pocket	Gage
108. Thickness of wall at base	Gage
109. Thickness of wall at counterbore end	Gage
110. Thickness of wall at datum location	Gage
111. Thickness of base at datum diameter	Gage
112. Burr	Visual
113. Foreign matter	Visual
MINOR	
201. Diameter of body at two locations, max.	Gage
202. Diameter of primer pocket	Gage
203. Diameter of primer pocket flash hole	Gage
204. Diameter of cut under flange	Gage
205. Diameter of body near base	Gage
206. Diameter of cavity below counterbore	Gage
207. Diameter of cavity	Gage
208. Depth of counterbore	Gage
209. Total length	Gage
210. Length of flange	Gage
211. Length of cut under flange	Gage
212. Length of chamfer	Gage
213. Radius or chamfer missing or incorrect	Visual
214. Surface finish improper	Visual

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4.3.2.3 Closure, Drawing C8594086, covering a detail of Drawing D8594084.

Defect		Method of Inspection
CRITICAL		
None defined		
MAJOR		
101.	Concentricity shoulder with body	Gage
102.	Diameter shoulder, min.	Gage
103.	Diameter of body, min.	Gage
104.	Depth of recess, min.	Gage
105.	Material defective	Visual
106.	Flash	Visual
107.	Foreign matter	Visual
MINOR		
201.	Diameter of shoulder, max.	Gage
202.	Diameter of body, max.	Gage
203.	Diameter of recess	Gage
204.	Diameter at chamfer	Gage
205.	Depth of recess, max.	Gage
206.	Length of body	Gage
207.	Total length	Gage
208.	Thickness through center	Gage
209.	Radius or chamfer missing or incorrect	Visual
210.	Surface finish improper	Visual

4.3.2.4 Seal, Primer-Vent, Drawing B8594087, covering a detail of Drawing D8594084.

CRITICAL		
None defined		
MAJOR		
101.	Diameter outside, min.	Gage
102.	Material defective	Visual
103.	Flash	Visual
104.	Foreign matter	Visual

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4.3.2.4 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Diameter outside, max.	Gage
202. Thickness	Gage

4.3.2.5 Assembly, Drawing D8594084.

CRITICAL	
1. Primer not properly below flush with surface "S"	Gage
2. Any part missing	Visual
3. Improperly assembled	Visual
4. Marking missing, incorrect or illegible	Visual
MAJOR	
101. Foreign matter	Visual
MINOR	
201. Diameter of body over marking, max.	Gage
202. Total length, max.	Gage
203. Varnish over marking missing or incorrect	Visual
204. Staking missing or incorrect	Visual

4.3.2.6 Ammunition Container, Metal M21 for Cartridge, Impulse, M57, Drawing F8796529.

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual/Visual
202. Quantity of contents incorrect	Visual
203. Separator or filler missing	Visual
204. Cut or damage through all impregnated layers	Visual
205. Marking missing, incorrect or illegible	Visual

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4.3.2.7 Box, Packing, Ammunition for Cartridge, Impulse M57 Assembly, Drawing C8796526.

Defect		Method of Inspection
CRITICAL		
None defined		
MAJOR		
None defined		
MINOR		
201.	Contents loose	Manual/Visual
202.	Hardware missing, loose, broken or improperly engaged	Visual
203.	Strapping missing, loose, broken or improperly assembled	Visual
204.	Contents of box exposed	Visual
205.	Box damaged	Visual
206.	Marking missing, incorrect or illegible	Visual
207.	Number of containers incorrect	Visual
208.	Car seal missing, unsealed or improperly positioned	Visual

4.3.3 Inspection testing.-The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Hardness.-Fifty cartridge cases from each heat-treated lot shall be tested. Failure of any unit to comply with the requirements of the drawing shall be cause for rejection of the lot.

4.3.3.2 Functional.-The following test shall be performed by the Government:

4.3.3.2.1 Ballistics.-Twenty-one cartridges from each lot shall be temperature conditioned at 70°F ± 5°F as prescribed in 4.4.2.1.1 and ballistically tested. Failure of any cartridge to comply with 3.2 shall be cause for rejection of the lot of cartridges.

4.3.4 Inspection equipment.-The examination and tests shall be made using equipment listed on IEL-8594084.

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4.4 Test methods and procedures.

4.4.1 Hardness.-A hardness testing machine shall be used to determine compliance with hardness requirement specified on the applicable drawing. The test shall be performed on the inside of the case in accordance with the drawing.

4.4.2 Functional.

4.4.2.1 Ballistics.-The test sample loaded in a test device shall be temperature conditioned and supported in the test fixture designed on the inspection equipment list. The sample shall be tested within 5 minutes after removal from the conditioning chamber. When this time limit is exceeded, the sample shall be reconditioned in accordance with 4.4.2.1.1.

4.4.2.1.1 Temperature conditioning.-A standard M6 Catapult, loaded with a dummy M57 Cartridge, containing a temperature sensitive element, shall be placed in the conditioning chamber at the same time as the samples. The conditioning time shall be a minimum of 1 hour longer than the time necessary to bring the element contained in the dummy cartridge to the required temperature.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.-Packing and marking shall be in accordance with Drawings F8796529 and C8796526.

6. Notes

6.1 Ordering data.-Procurement documents should specify the title, number and date of this specification.

CUSTODIAN:

Army - MU

PREPARING ACTIVITY:

Army - MU(FA)

Project No. 1377-A484

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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